

AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science

Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

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Section: [*L*]

Group No: 00

Project Report On

Beauty Salon Management System

Supervised By

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Chapter :01 (Introduction)

Introduction:

The **Beauty Salon Management System** is an easy-to-use application made to help salon owners and staff manage their daily tasks. It brings all the important features into one system—such as booking appointments, managing services, keeping track of products, handling payments, and storing customer and staff information.

This system helps avoid mistakes, saves time, and keeps everything well-organized. Customers can book appointments without visiting the salon, and staff can easily check schedules and service lists. With this system, the salon becomes professional, faster, and better at meeting customer needs.

It also improves communication between customers and salon workers, helping to provide better service. The system stores all data safely, so nothing gets lost, and it makes it easier for everyone to do their jobs.

Objective: The main goal of the Beauty Salon Management System is to make salon work easier and more organized by using technology. Here's what the system will achieve:

- Help customers **book appointments online** without needing to call or visit.
- Allow staff to **manage their own profiles**, work schedules, and services.
- Let the admin easily **add, edit, or remove** staff, services, categories, and products.
- Keep track of **product stock and prices** to avoid shortages.
- Generate **accurate bills** and let customers pay by cash, card, or mobile payment (like bKash or Nagad).
- Show useful information on a **dashboard** for Admins and Staff to see appointments, sales, and updates.
- Let customers **choose their preferred service and time** when booking.
- Make sure all user data is stored **securely** and only the right person can access their account.

Scope:

The system will be used by three types of users:

- **Admin** – Can manage staff, services, categories, products, appointments, and bills.
- **Staff** – Can assist customers, manage services/products, book appointments, and handle payments.
- **Customer** – Can create an account, book appointments, choose services, and view their bills

Main Features:

- User login, signup, and password reset
- Appointment booking system
- Manage services, categories, and products
- Payment system with cash, cards, or mobile options
- Bill generation and report viewing
- Admin and staff dashboard
- Easy profile management for all users

Tools Used:

- **Programming Language:** C#
- **Database:** SQL Server Management studio
- **IDE:** Visual Studio

Development Approach:

The system is built using **Object-Oriented Programming (OOP)**. This helps keep the code clean and organized.

- **Encapsulation:** Keeps user and data safe inside the system.
- **Inheritance:** Allows code reused (Admin and Staff both share user features).
- **Abstraction:** Shows only necessary details to users and hides complex parts.
- **Polymorphism:** Lets different users (Admin, Staff, Customer) use the same features in their own way.

Chapter :02 (User Story)

User1: Admin

The admin has access to key management features to efficiently manage staff, products, categories, services, and orders. This includes updating staff credentials, managing product details, organizing categories, and monitoring the balance sheet.

1.User management:

- Admins can easily update staff details such as their username, email, address, contact information, and salary.
- The system will ensure staff can log in and out securely, keeping their information safe.
- Admins will also be able to add new staff or remove existing ones whenever needed.
- There will be an option to clear staff data, making it easier for admins to manage and update information when required.
    - Admin should be able to update his profile like updating name, address, phone No, email

2. Product Management:

- Admin should be able to manage product details like ID, name, category, status, unit, price, and other important information.
- The system needs to make sure product data is securely stored and kept safe.
- Admin should be able to delete products that are no longer needed or useful.
- A clear option should be available to reset services information fields.

3. Category Management:

- Admin should have the ability to easily search for and view existing categories in the system.
- Adding new categories should be a quick and simple task for the admin.
- The system should allow for the removal of categories when required.
- Admin should be able to clear category data fields for smoother management.

4. Bill Management:

- Admin should be able to easily check a balance sheet to keep track of all products and quantities, making sure everything is accurate, and inventory is managed well.
- Admin should also be able to review customer orders for new services and create bills for those orders, giving customers the choice to pay with cash or card.

User2: Staff

1.User management:

- Staff should be able to update their profile, including changes to their name, address, phone number, and email.
- The system must ensure that profile information is securely saved.
- Staff should be able to delete their account, search for specific information, change their password, and clear data fields when necessary.

2. Product Management:

- Staff should be able to manage products including their ID, Name, status, unit, price, etc.
- The system should allow saving product information securely.
- Staff should have the capability to delete products that are no longer needed. - A clear option should be available to reset product information.

3. Service Management: -

- Staff should be able to manage services, including updating their name, price, status, and other important details.
- The system should ensure that service information is securely stored.
- Staff should have the ability to remove services that are no longer required.
- An option to clear service information fields should be available for better management.

4. Category Management:

- Adding new categories should be a straightforward process for staffs.
- The system should support the deletion of categories when necessary.
- Staff should have the option to clear category data fields.

5.Bill Management:

- Staff can receive payment.
- Staff can return cashback for customer.
- Customer must receive a bill token after completing his/her payment where mention how many items and quantity customer orders and also show billing date, time, customer name.

6.Give Appointment:

- If the customer contacts the staff for an appointment, the staff will book the appointment for the customer.

User3: Customer

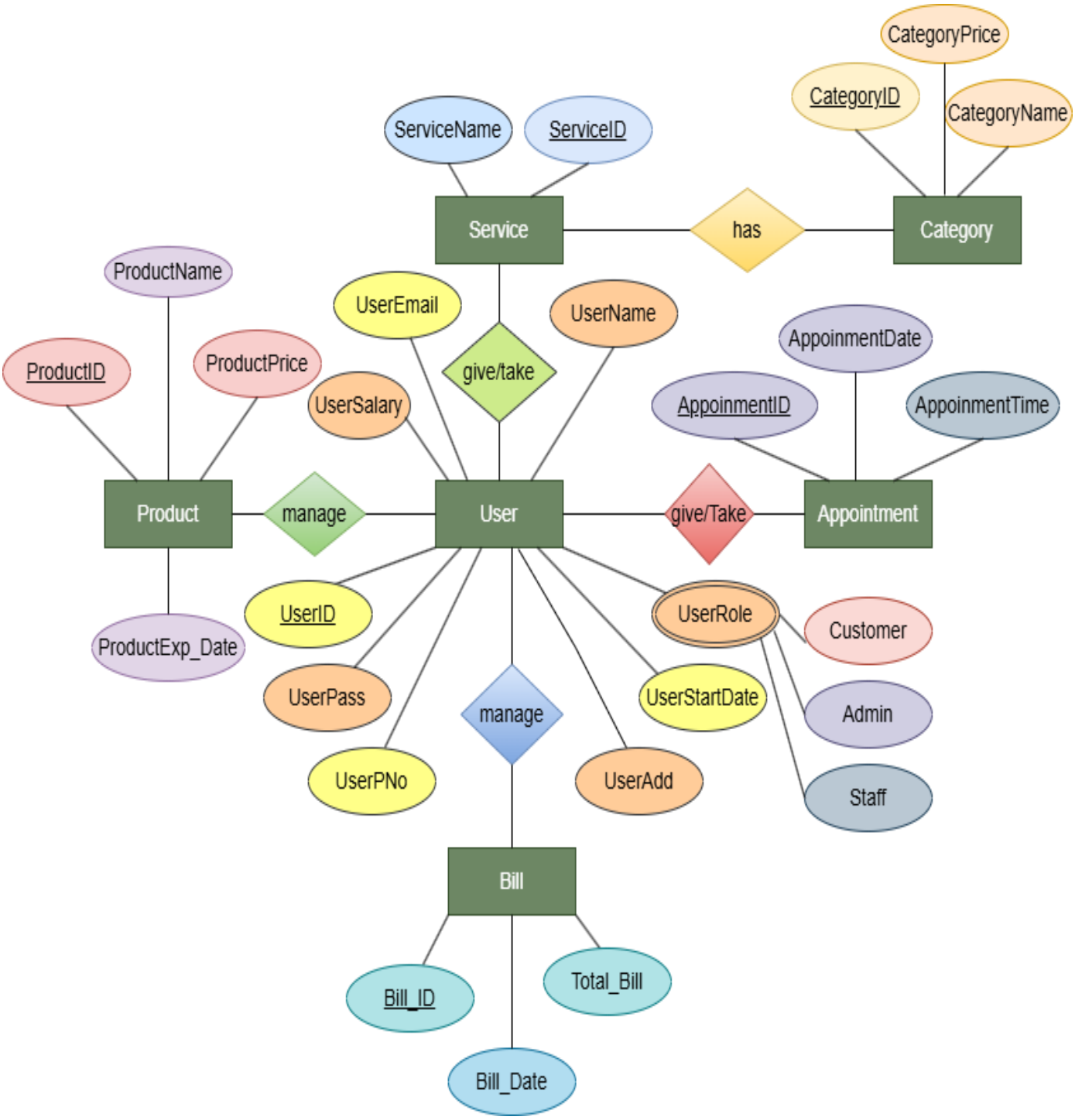
1.Take appointment:

- The customer will be able to book an appointment and view the details of their appointment.
- When making the appointment, the customer will be able to choose the service and category according to their preference.
- Then, at the scheduled time, the customer will be able to receive the selected service.

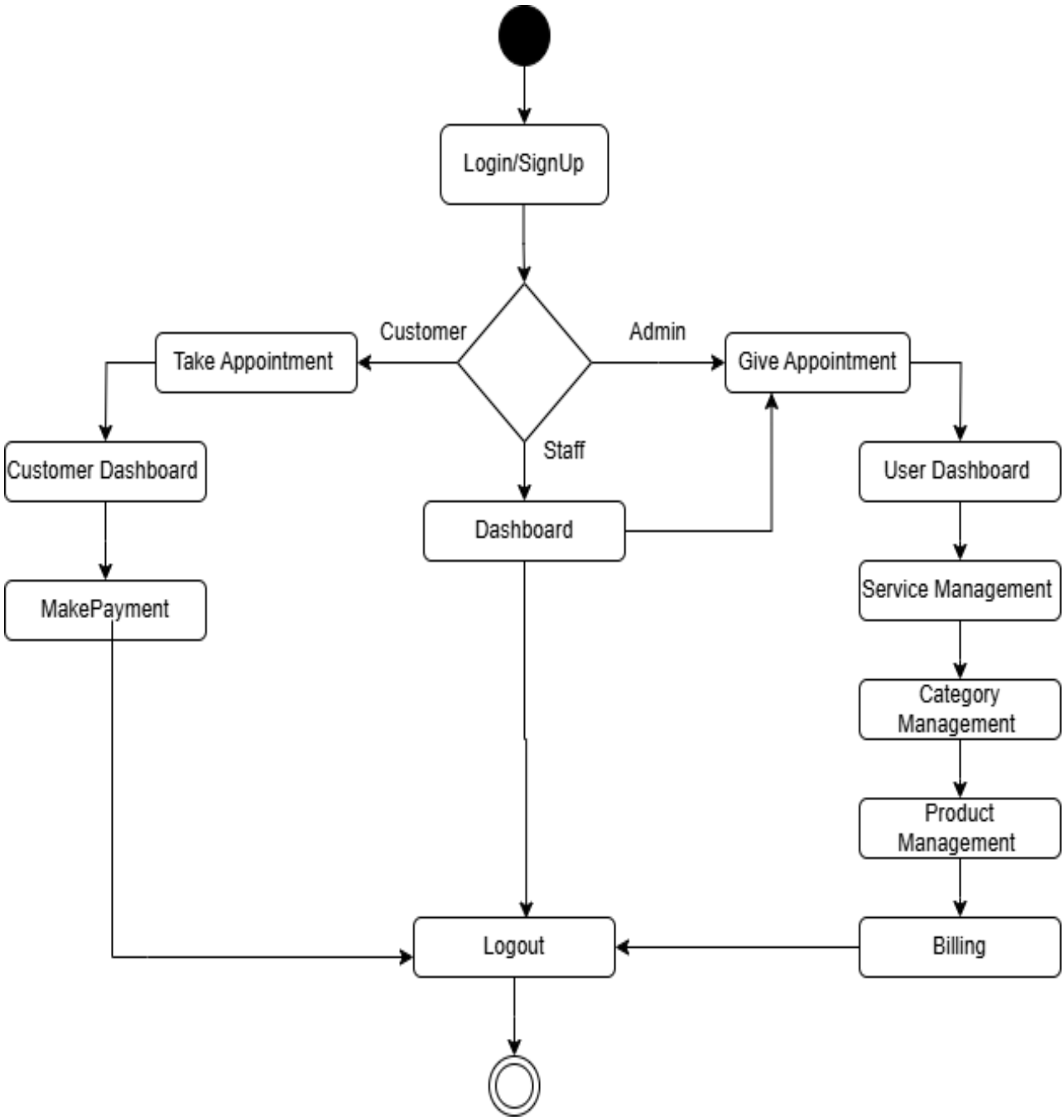
SYSTEM OVERVIEW AND WORKFLOW

- The system will start with a login page where **customers**, **admins,** and **staff** can login using their respective IDs and passwords. The login page will also include options for sign-up and password recovery. If any user forgets their password, they can reset it by providing their user ID and phone number.
- If the user is a customer, they will be able to book appointments after logging in. Customers can book an appointment by entering their ID, date, and time. Additionally, during the booking process, they can select their preferred service and category.
- On the other hand, admins and staff will also use the same login page to access their accounts. After logging in, they will be directed to a dashboard with various modules. Through the dashboard, admins and staff can view all customer appointment details and billing information. The user management module will contain all the details of staff and admins. The product management module will display the shop's entire product list, including product names, prices, and stock information. The service management module will include information about all available services, while the category management module will list all service categories.
- In the billing module, the total bill will be calculated based on the products or services selected by the customer and their respective quantities. Customers can make payments using cash or card, and additionally, they will have the option to pay online through bKash, Nagad, or similar mobile financial services. The entire system is designed to be user-friendly, ensuring smooth functionality for customers, admins, and staff.

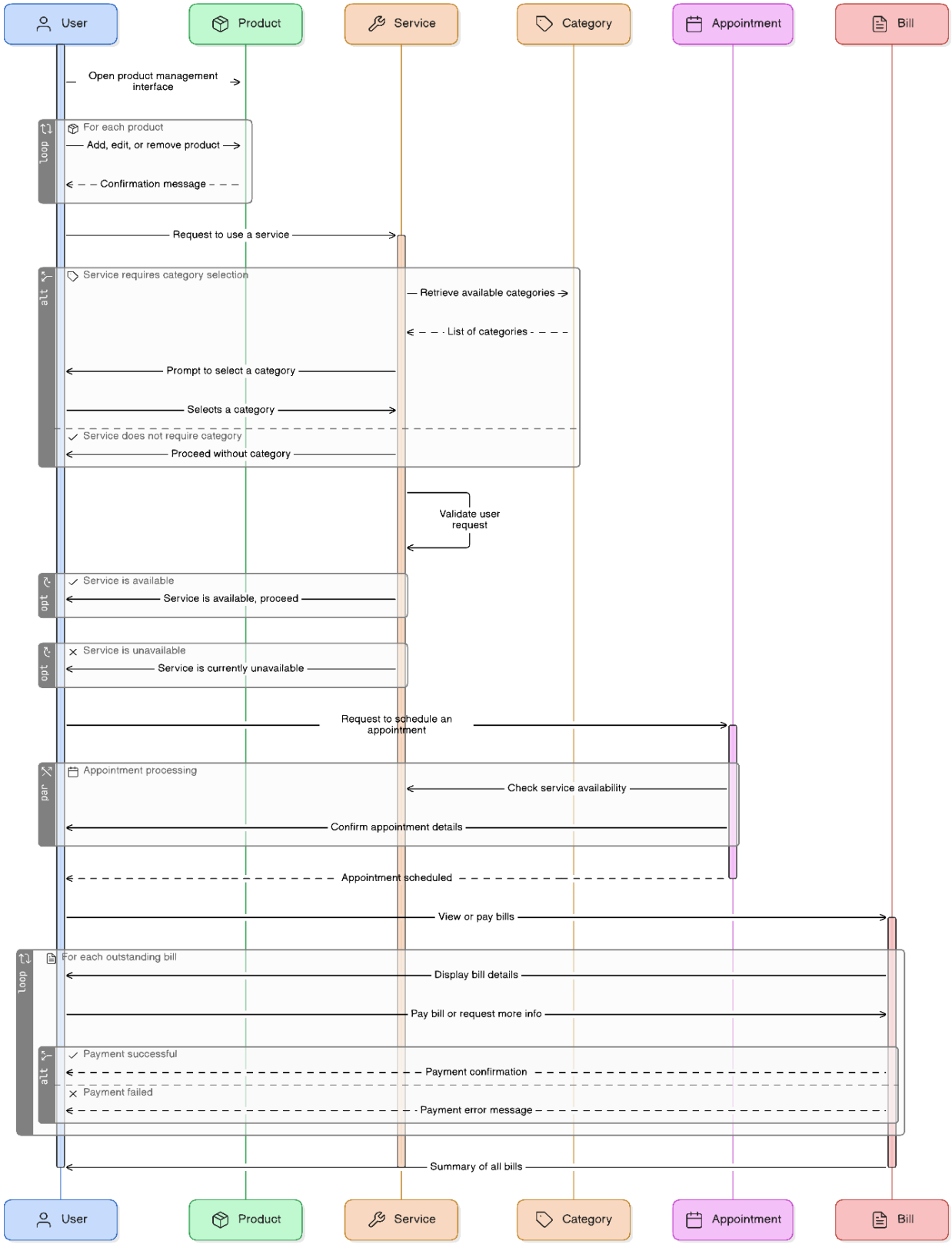
Chapter :03 (ER-DIAGRAM)



ACTIVITY DIAGRAM



SEQUENCE DIAGRAM



NORMALIZATION:

1.User → Product (Manage)

UNF :

UserID, UserRole, UserAddress, UserName, UserSalary,UserPassword,UserEmail,

UserStartDate, UserPhonenum, ProductID, ProductPrice, ProductExpdate, ProductName.

1NF :

- UserID,UserRole,UserAddress,UserName,UserSalary,UserPassword,UserEmail, UserStartDate, UserPhonenum
- ProductID, ProductPrice, ProductExpdate, ProductName.

2NF :

- UserID,UserRole,UserAddress,Username,UserSalary,UserPassword,UserEmail,UserStartDate, UserPhonenum.
- ProductID(PK), ProductPrice ,ProductExprdate, ProductName,UserID(FK).

3NF:

- No transitive dependency

2.User → Service(Give/take)

UNF :

UserID, UserRole, UserAddress, UserName, UserSalary,UserPassword,UserEmail,

UserStartDate, UserPhonenum , ServiceID, ServiceName.

1NF :

- UserID, UserRole, UserAddress, UserName, UserSalary,UserPassword,UserEmail,UserStartDate, UserPhonenum .
- ServiceID, ServiceName.

2NF :

- UserID,UserRole,UserAddress,UserName,UserSalary,UserPassword,UserEmail,UserStartDate, UserPhonenum
- ServiceID(PK), ServiceName, UserID(FK).

3NF:

- No transitive dependency

3.Service→ Category(has)

UNF :

CategoryID, CategoryName,CategoryPrice, ServiceID, ServiceID Name

1NF :

- CategoryID, CategoryName,CategoryPrice
- ServiceID, ServiceName.

2NF:

- CategoryID, CategoryName,CategoryPrice,
- CategoryID(PK), CategoryName,CategoryPrice, ServiceID(FK)

3NF:

- No transitive dependency

4.User → Bill(manage)

UNF :

UserID, UserRole, UserAddress, UserName, UserSalary,UserPassword,UserEmail,

UserStartDate, UserPhonenum,BillName, BillID, TotalBill.

1NF:

- UserID, UserRole, UserAddress, UserName, UserSalary,UserPassword,UserEmail, UserStartDate
- BillName, BillID, TotalBill.

2NF:

- UserID,UserRole,UserAddress,UserName,UserSalary,UserPassword,UserEmail,UserStart Date,UserPhonenum.
- BillID(PK), BillID, TotalBill,UserID(FK).

3NF:

- No transitive dependency

5.User → Appointment(give/take)

UNF:

UserID, UserRole, UserAddress, UserName, UserSalary,UserPassword,UserEmail,

UserStartDate, UserPhonenum,AppoinmentID, AppoinmentDate, AppoinmentTime.

1NF:

- UserID, UserRole, UserAddress, UserName, UserSalary,UserPassword,UserEmail, UserStartDate, UserPhonenum.
- AppoinmentID, AppoinmentDate, AppoinmentTime.

2NF:

- UserID, UserRole, UserAddress, UserName, UserSalary,UserPassword,UserEmail, UserStartDate, UserPhonenum.
- AppoinmentID(PK), AppoinmentDate, AppoinmentTime, UserID(FK)

3NF:

- No transitive dependency

Chapter :03 (SQL Queries)

SQL Tables:

Column Name	Data Type	Allow Nulls
ProductID	int	☐
ProductName	nvarchar(50)	☑
ProductPrice	nvarchar(50)	☑
ProductExp_Date	nvarchar(50)	☑

Column Name	Data Type	Allow Nulls
BillingID	int	☑
CategoryID	nvarchar(50)	☑
ProductID	nvarchar(50)	☑
TotalBill	nvarchar(50)	☑
CustomerID	nvarchar(50)	☑
BillingDate	nvarchar(50)	☑

Column Name	Data Type	Allow Nulls
ServiceID	int	☑
ServiceName	nvarchar(50)	☑

Column Name	Data Type	Allow Nulls
CategoryID	int	☑
CategoryName	nvarchar(50)	☑
CategoryPrice	int	☑
ServiceID	int	☑

	Column Name	Data Type	Allow Nulls
▶	UserID	nvarchar(50)	☐
	UserName	nvarchar(50)	☑
	UserPass	nvarchar(50)	☑
	UserPNo	nvarchar(50)	☑
	UserAdd	nvarchar(50)	☑
	UserStartDate	nvarchar(50)	☑
	UserSalary	nvarchar(50)	☑
	UserEmail	nvarchar(50)	☑
	UserRole	nvarchar(50)	☑

Column Name	Data Type	Allow Nulls
CHName	nvarchar(50)	☑
PhoneNo	nvarchar(50)	☑
Pin	int	☑
CardNo	varchar(50)	☑

	Column Name	Data Type	Allow Nulls
	AppointmentID	int	☑
	AppointmentDate	nvarchar(50)	☑
	AppointmentTime	nvarchar(50)	☑
	ServiceName	nvarchar(50)	☑
	CategoryName	nvarchar(50)	☑
	CustomerID	int	☑
▶			☐
			☐

SQL Table Creation Query:

1.User Table:

```
CREATE TABLE [dbo].[userInfo](
	[UserID] [int] NULL,
	[UserName] [nvarchar](50) NULL,
	[UserPass] [nvarchar](50) NULL,
	[UserPNo] [nvarchar](50) NULL,
	[UserAdd] [nvarchar](50) NULL,
	[UserStartDate] [nvarchar](50) NULL,
	[UserSalary] [nvarchar](50) NULL,
	[UserEmail] [nvarchar](50) NULL,
	[UserRole] [nvarchar](50) NULL
) ON [PRIMARY]
```

2.Appointment Table:

```
 CREATE TABLE [dbo].[Appointment](
	[AppointmentID] [int] NULL,
	[AppointmentDate] [nvarchar](50) NULL,
	[AppointmentTime] [nvarchar](50) NULL,
	[ServiceName] [nvarchar](50) NULL,
	[CategoryName] [nvarchar](50) NULL,
	[CustomerID] [int] NULL
) ON [PRIMARY]
```

3.Product Table:

```
 CREATE TABLE [dbo].[Product](
	[ProductID] [int] NULL,
	[ProductName] [nvarchar](50) NULL,
	[ProductPrice] [nvarchar](50) NULL,
	[ProductExp_Date] [nvarchar](50) NULL
) ON [PRIMARY]
```

4.Category Table:

```
  CREATE TABLE [dbo].[Category](
	[CategoryID] [int] NULL,
	[CategoryName] [nvarchar](50) NULL,
	[CategoryPrice] [int] NULL,
	[ServiceID] [int] NULL
) ON [PRIMARY]
```

5.Payment Table:

```
CREATE TABLE [dbo].[Payment](
	[CHName] [nvarchar](50) NULL,
	[PhoneNo] [nvarchar](50) NULL,
	[Pin] [int] NULL,
	[CardNo] [nvarchar](50) NULL
) ON [PRIMARY]
```

6.Billing Table:

```
CREATE TABLE [dbo].[Billing](
	[BillingID] [int] NULL,
	[CategoryID] [int] NULL,
	[ProductID] [int] NULL,
	[TotalBill] [int] NULL,
	[CustomerID] [int] NULL,
	[BillingDate] [nvarchar](50) NULL
) ON [PRIMARY]
```

7.Service Table:

```
CREATE TABLE [dbo].[Service](
	[ServiceID] [int] NULL,
	[ServiceName] [nvarchar](50) NULL
) ON [PRIMARY]
```

SQL Insert Queries

1. **UserInfo:** INSERT INTO userInfo (UserID, UserName, UserPass, UserPNo, UserAdd, UserStartDate, UserSalary, UserEmail, UserRole)

VALUES (@UserID, @UserName, @UserPass, @UserPNo, @UserAdd, @UserStartDate, @UserSalary, @UserEmail, @UserRole);

2. **Product:** INSERT INTO Product (ProductID, ProductName, ProductPrice, ProductExp_Date)

VALUES (@ProductID, @ProductName, @ProductPrice, @ProductExp_Date);

3. Service: INSERT INTO Service (ServiceID, ServiceName)

VALUES (@ServiceID, @ServiceName);

4. Category: INSERT INTO Category (CategoryID, CategoryName)

VALUES (@CategoryID, @CategoryName);

SQL Select Queries

1. **UserInfo:**

SELECT UserRole FROM userInfo WHERE UserID = @UserID AND UserPass COLLATE SQL_Latin1_General_CP1_CS_AS = @UserPass;

SELECT UserID, UserName, UserPNo, UserAdd, UserEmail, UserRole FROM userInfo WHERE UserID = @UserID;

SELECT UserRole FROM userInfo WHERE UserID = @UserID;

SELECT COUNT(UserID) AS totalStaff FROM userInfo WHERE UserRole = 'Staff';

2. **Product:**

SELECT * FROM Product;

SELECT * FROM Product WHERE ProductID LIKE @searchTerm OR ProductName LIKE @searchTerm

OR ProductPrice LIKE @searchTerm;

3. Service:

SELECT * FROM Service;

SELECT ServiceID, ServiceName FROM Service WHERE ServiceID = @ServiceID;

4.Payment Table:

SELECT COUNT(*) FROM Payment WHERE PhoneNo = @PhoneNo AND Pin = @Pin;

SELECT COUNT(1) FROM Payment WHERE CHName = @CHName AND CardNo = @CardNo;

5.Billing Table:

SELECT * FROM Billing;

SELECT * FROM Billing WHERE CustomerID LIKE @searchTerm;

6.Category: SELECT * FROM Category;

7.Appointment: SELECT * FROM Appointment;

SQL Update Queries

1. **UserInfo:**

UPDATE userInfo

SET UserName = @UserName, UserPNo = @UserPNo, UserAdd = @UserAdd, UserEmail = @UserEmail

WHERE UserID = @UserID;

UPDATE userInfo SET UserPass = @NewPassword WHERE UserID = @UserID;

2. **Product:**

UPDATE Product

SET ProductName = @ProductName, ProductPrice = @ProductPrice, ProductExp_Date = @ProductExp_Date

WHERE ProductID = @ProductID;

3. Service:

UPDATE Service

SET ServiceName = @ServiceName

WHERE ServiceID = @ServiceID;

3.Category:

UPDATE Category SET CategoryName = @CategoryName WHERE CategoryID = @CategoryID;

SQL Delete Queries

1.UserInfo: DELETE FROM userInfo WHERE UserID = @UserID;

2. Product: DELETE FROM Product WHERE ProductID = @ProductID;

3.Service: DELETE FROM Service WHERE ServiceID = @ServiceID;

4.Category: DELETE FROM Category WHERE CategoryID = @CategoryID;

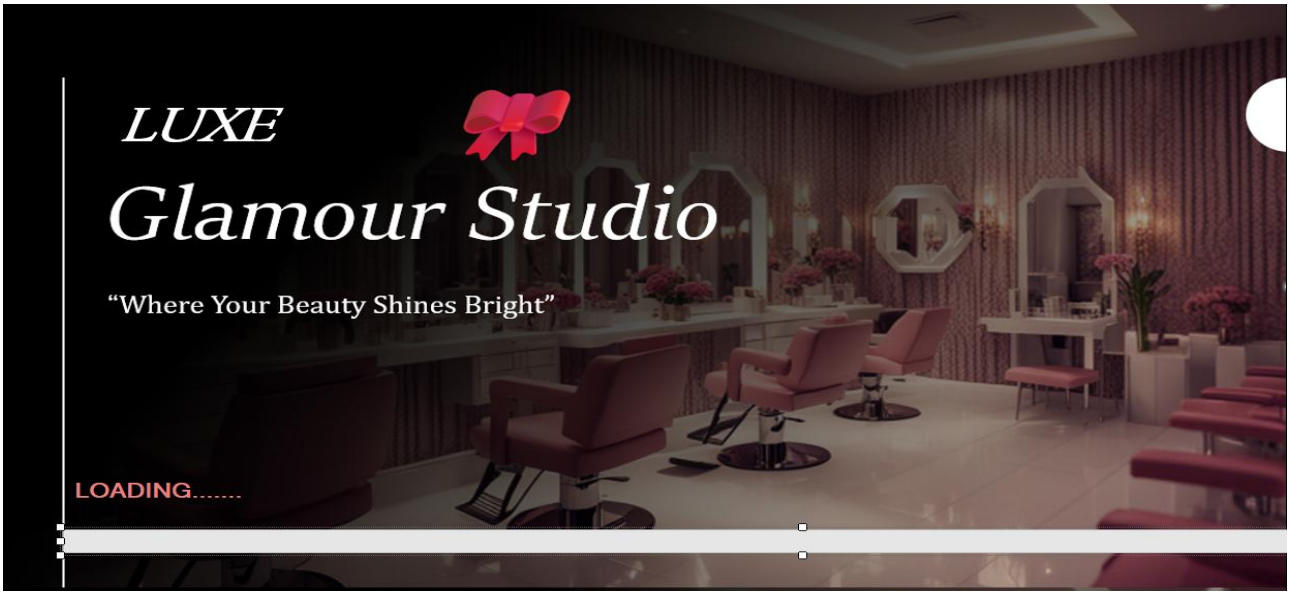
SQL Search Queries

1.Service: SELECT * FROM Service WHERE ServiceName LIKE @searchTerm OR ServiceID LIKE @searchTerm;

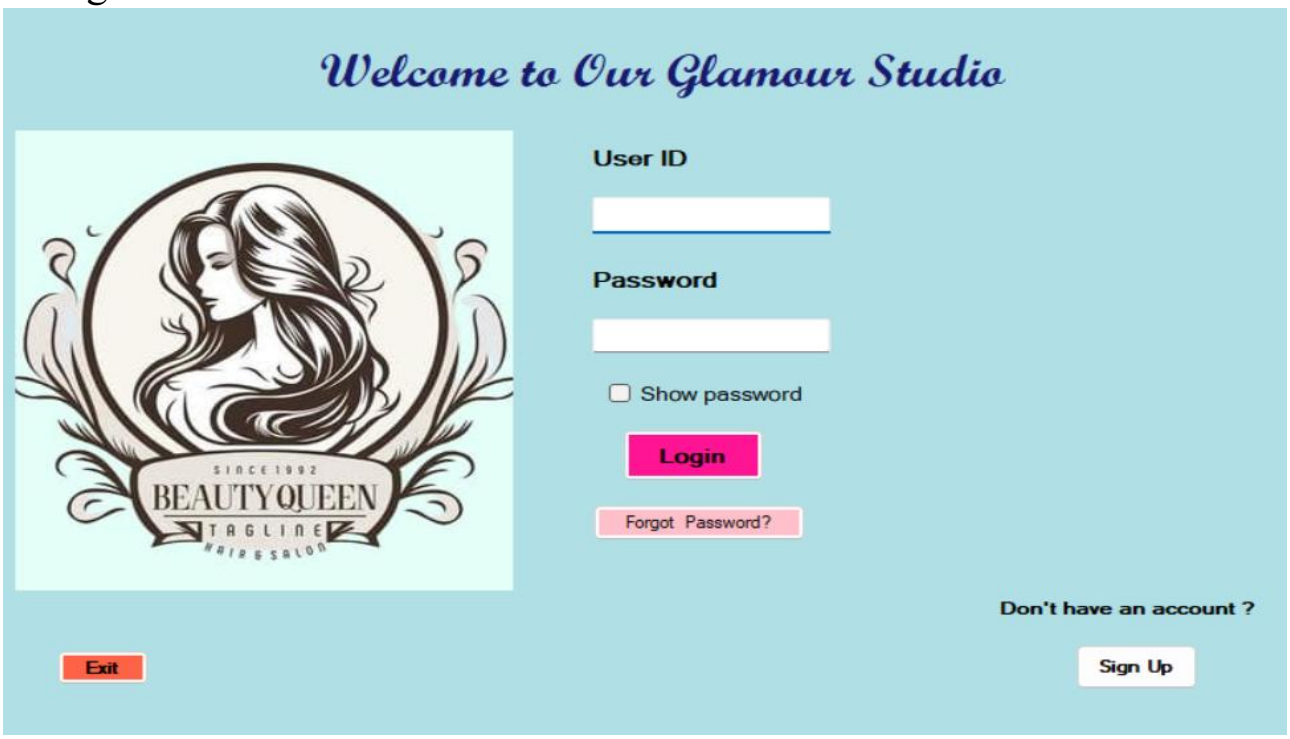
2.Category: SELECT * FROM Category WHERE CategoryName LIKE @searchTerm OR CategoryID LIKE @searchTerm;

Chapter: 05 (SCREENSHORTS)

1.Start loading:



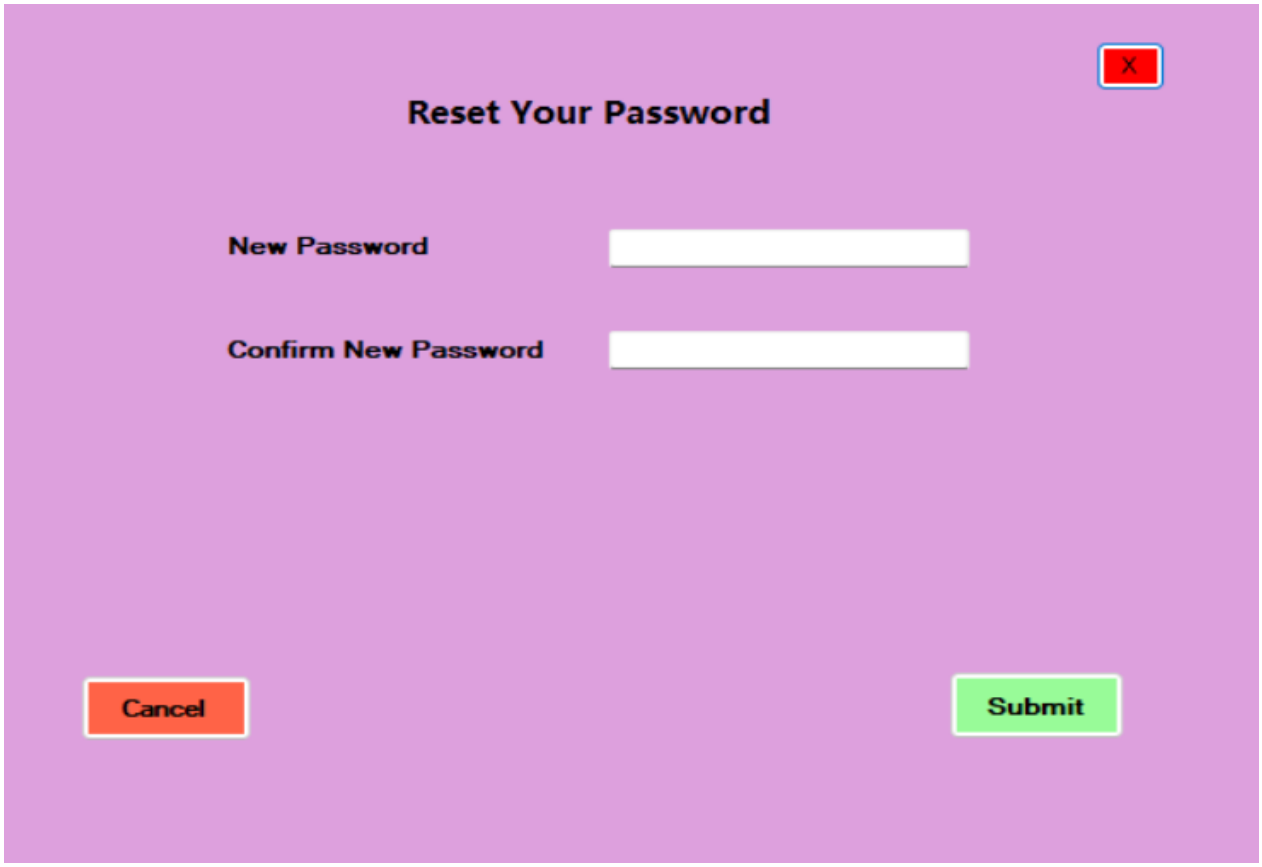
2. Login Window:



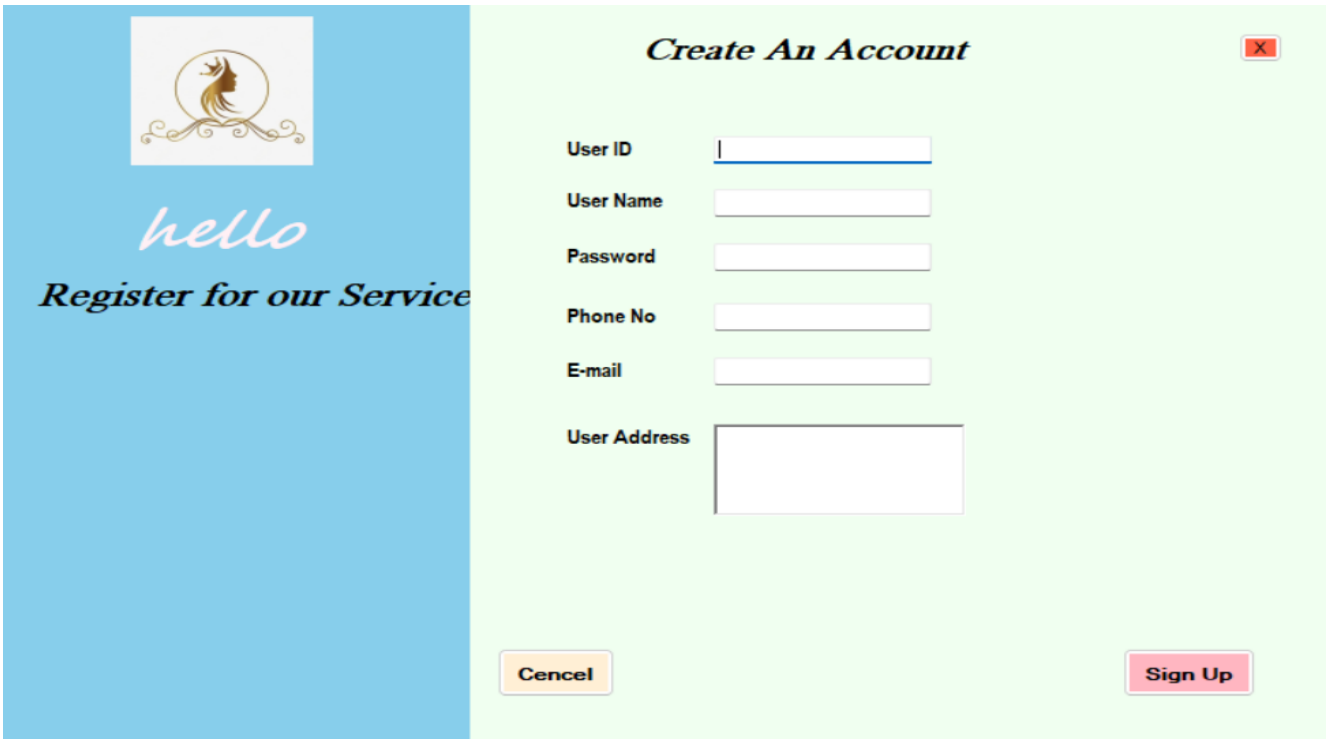
3.Forgot Password:



4.Reset Password:



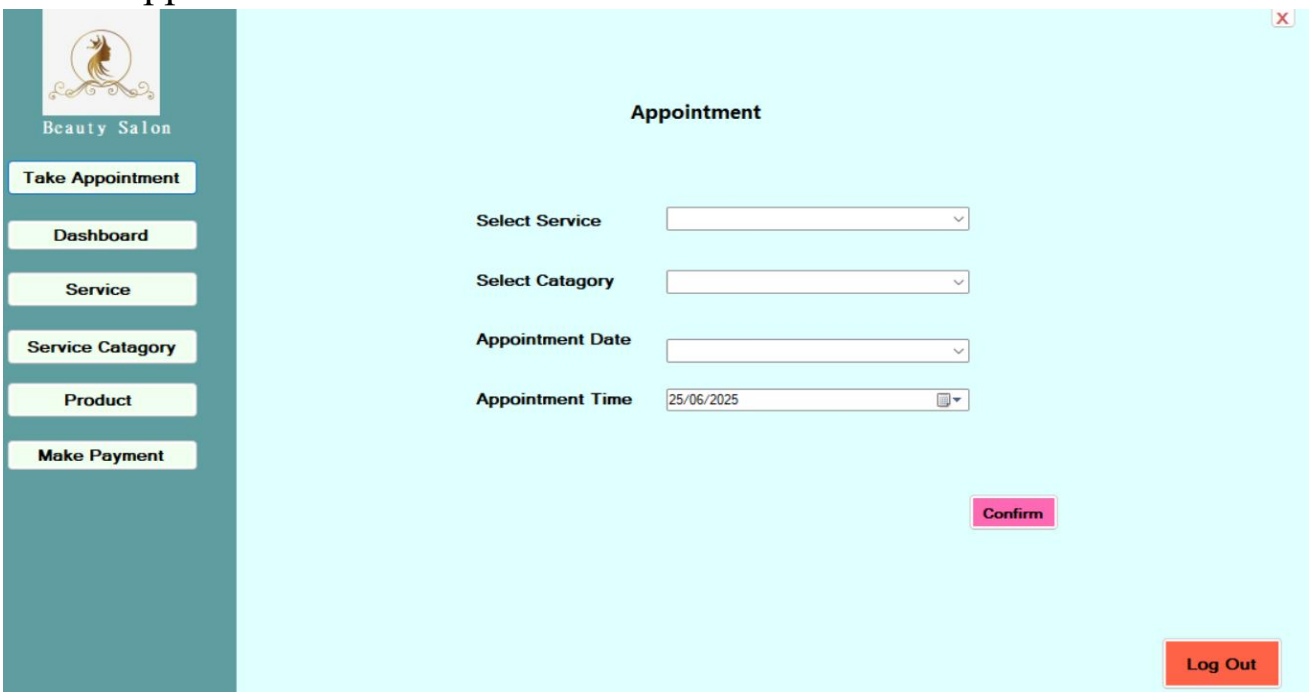
5.Sign Up



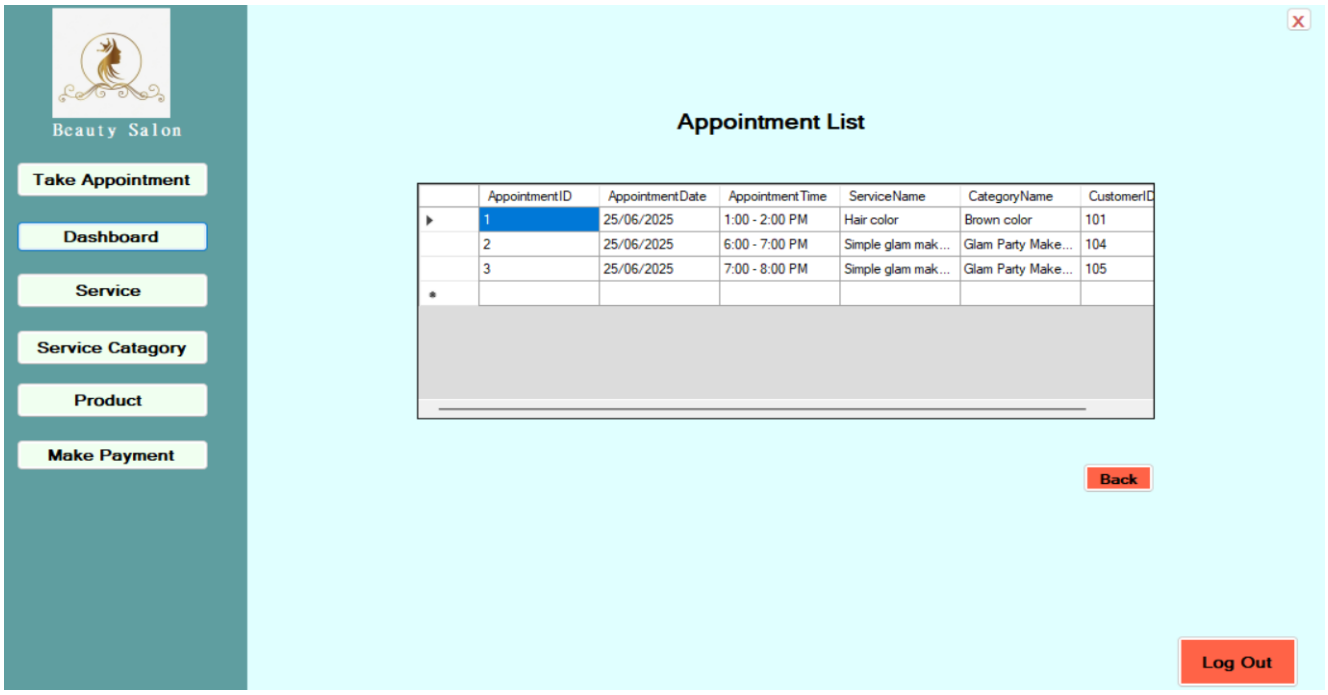
6.Customer Window:



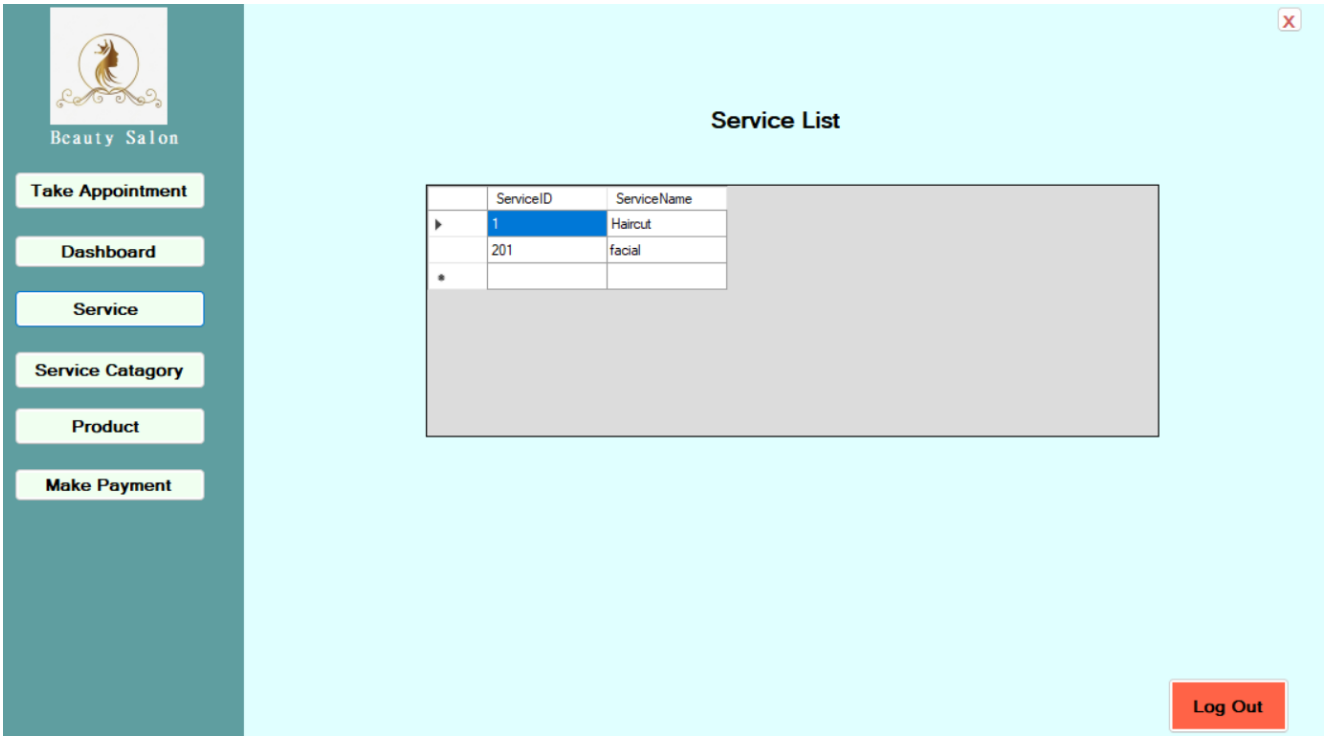
7.Take Appointment:



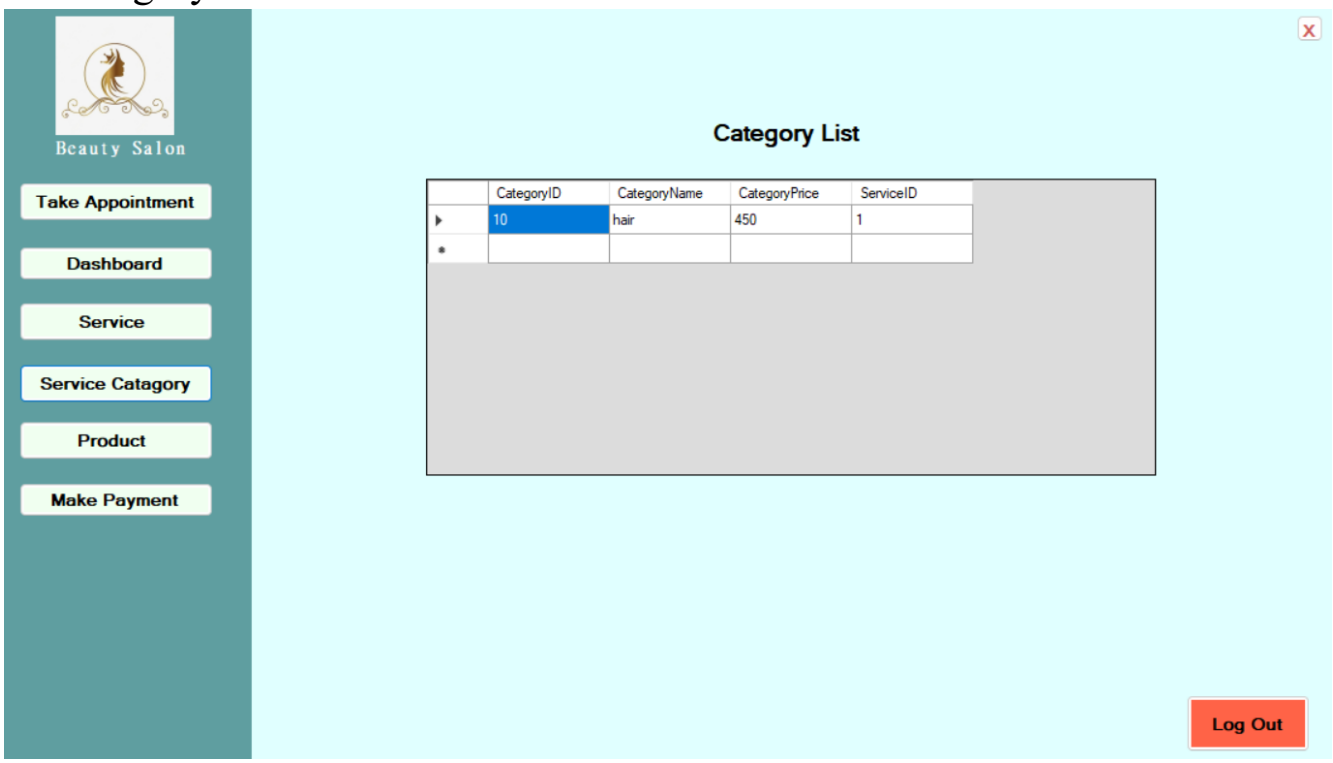
8.Customer Dashboard:



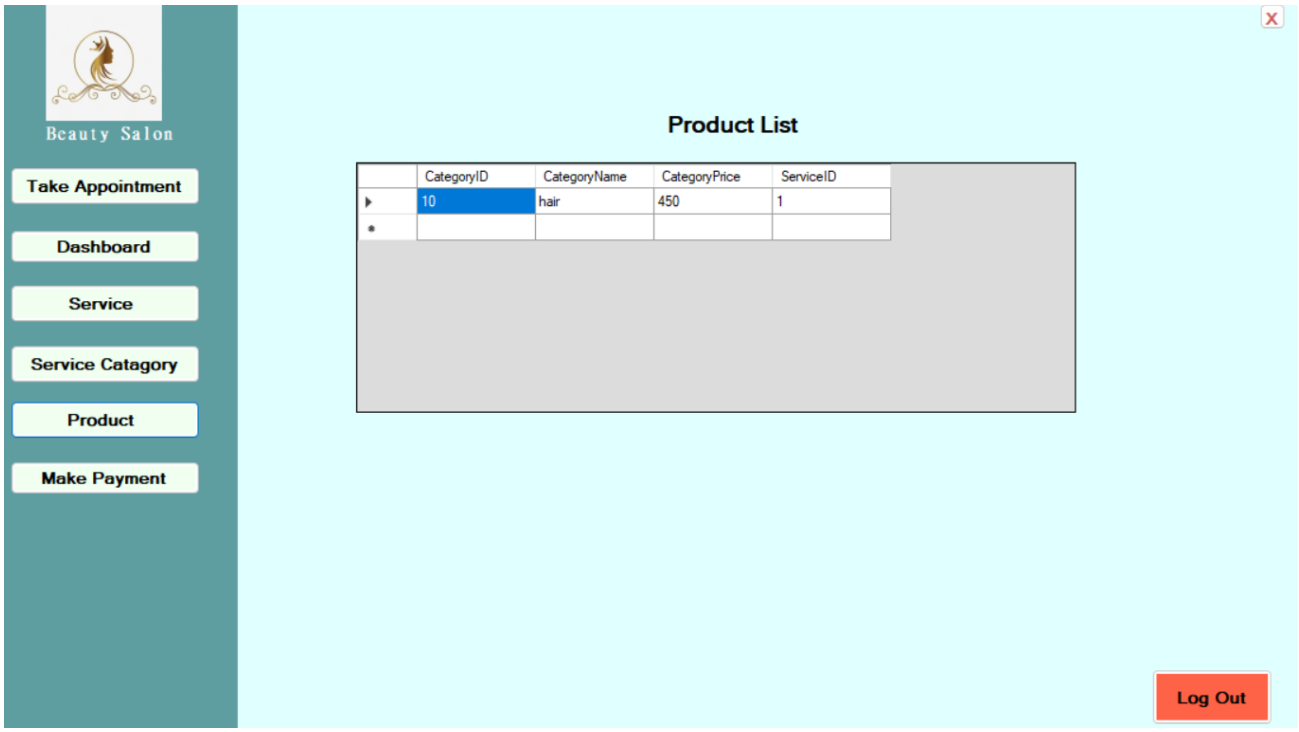
9. Service List:



10.Category:

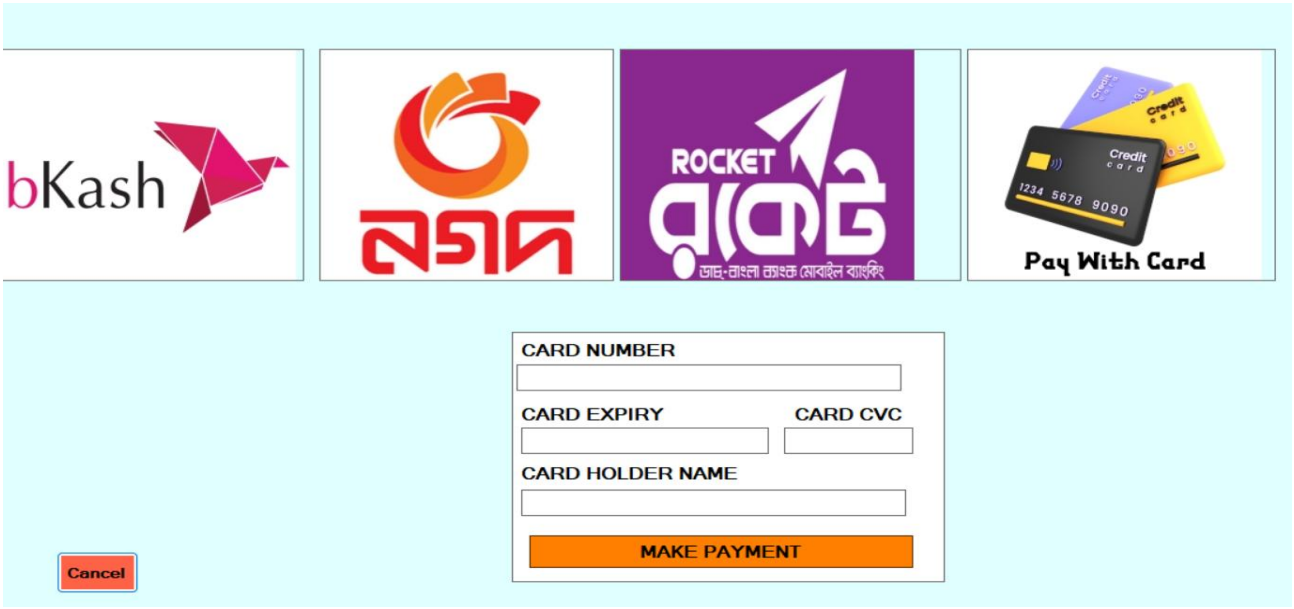


11.Product List:



12.Make Payment:

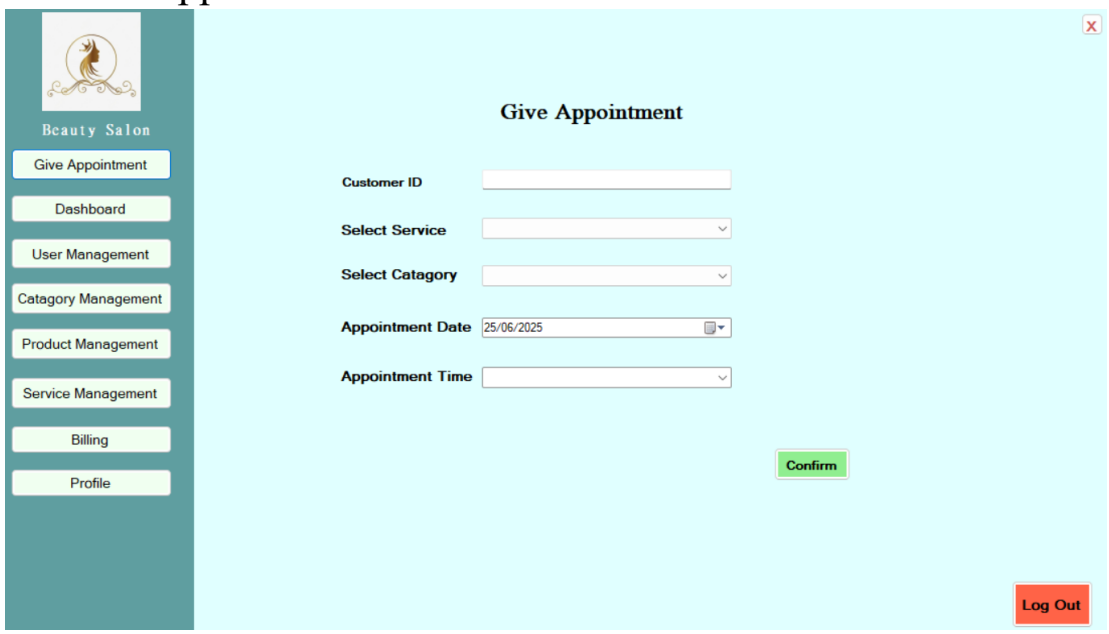




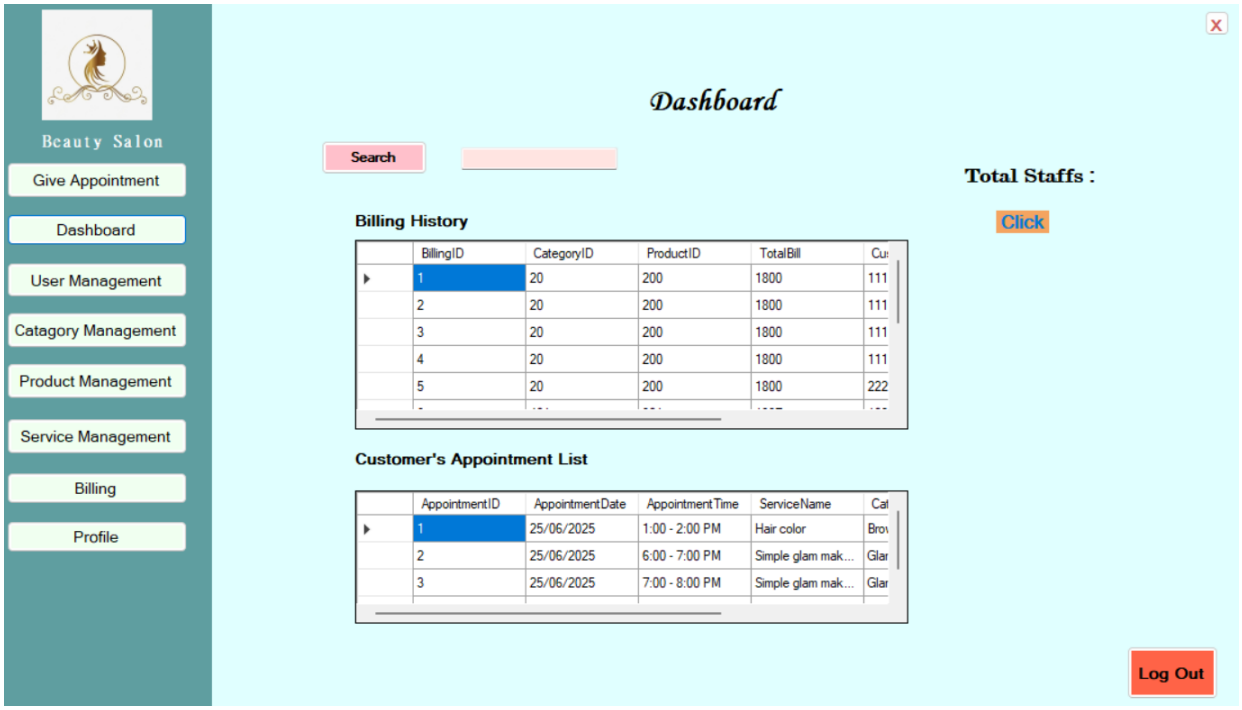
13.Admin/Staff Window:



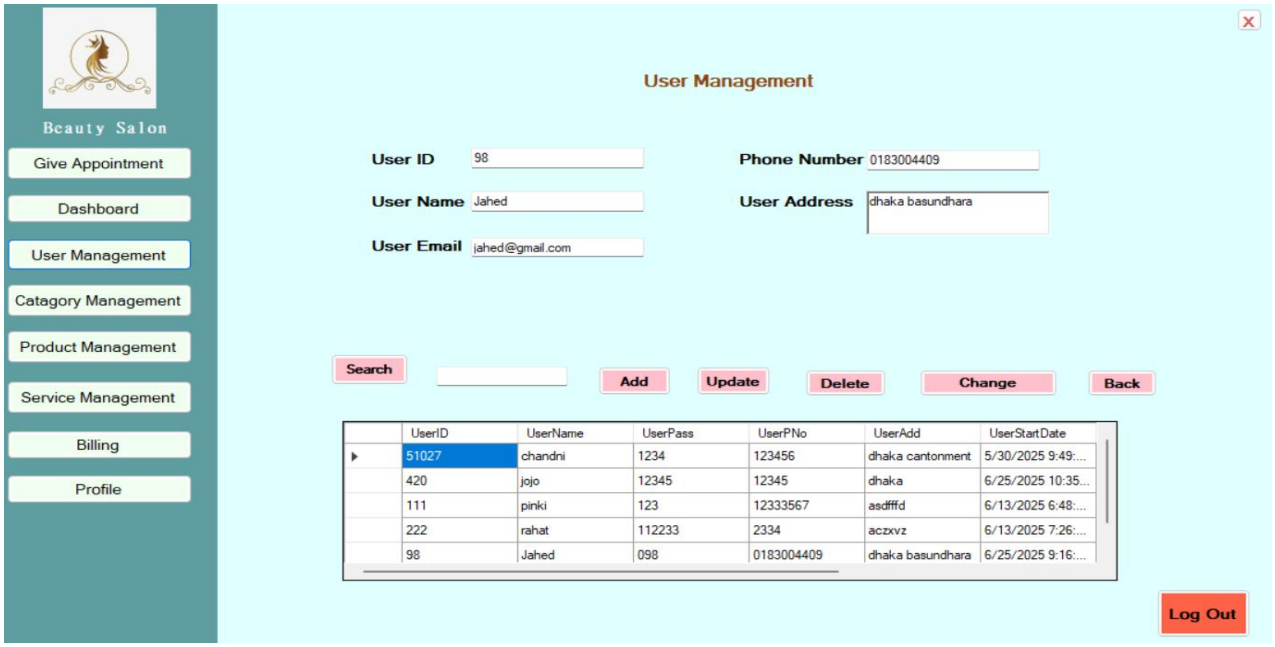
14. Give Appointment:



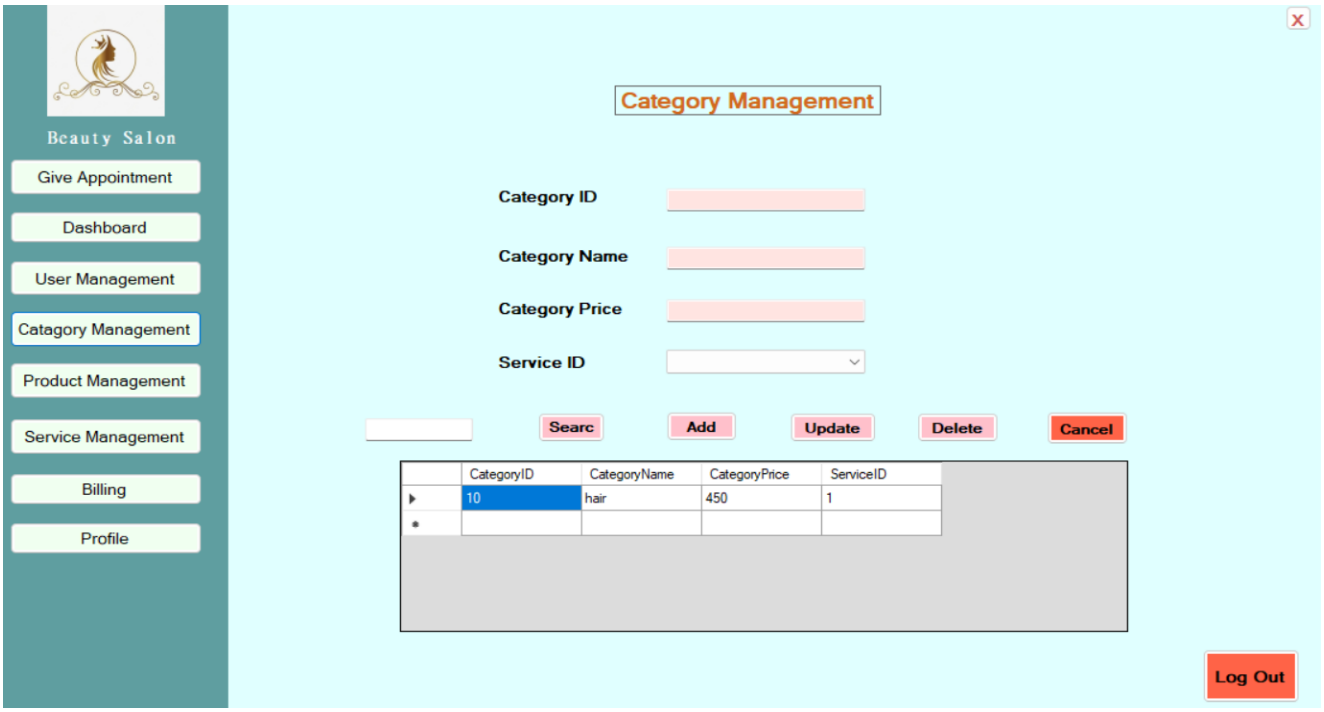
15.DashBoard:



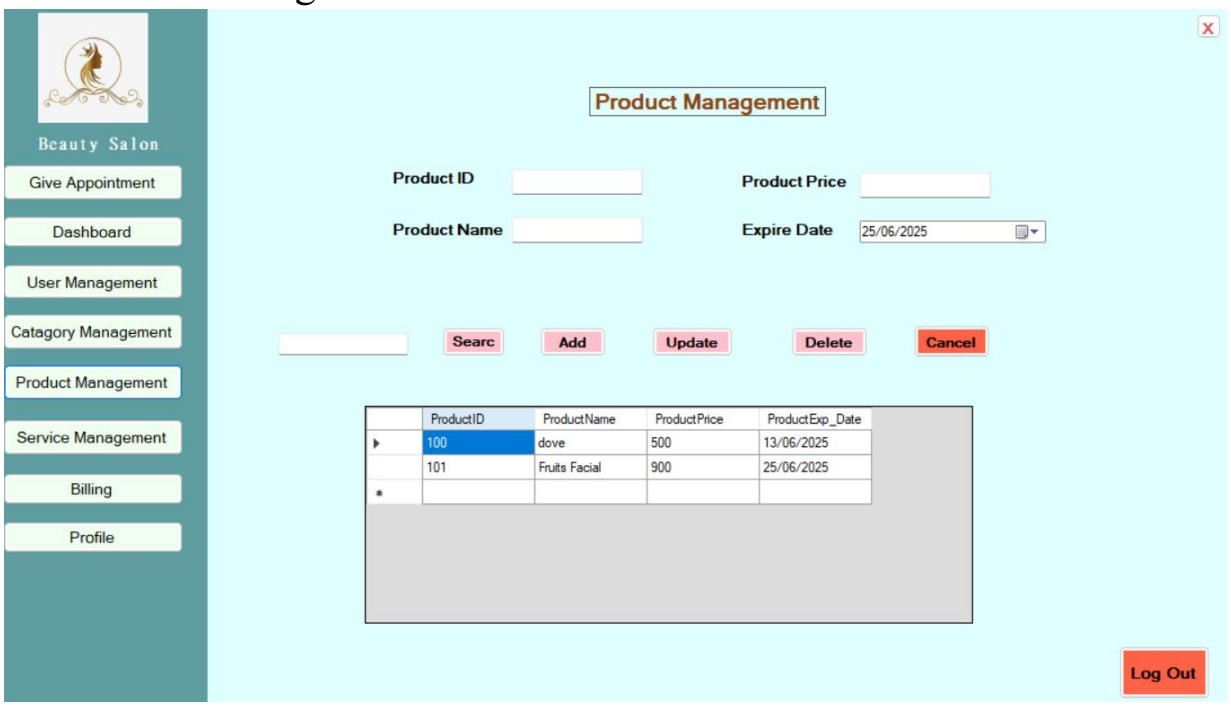
16.User Management:



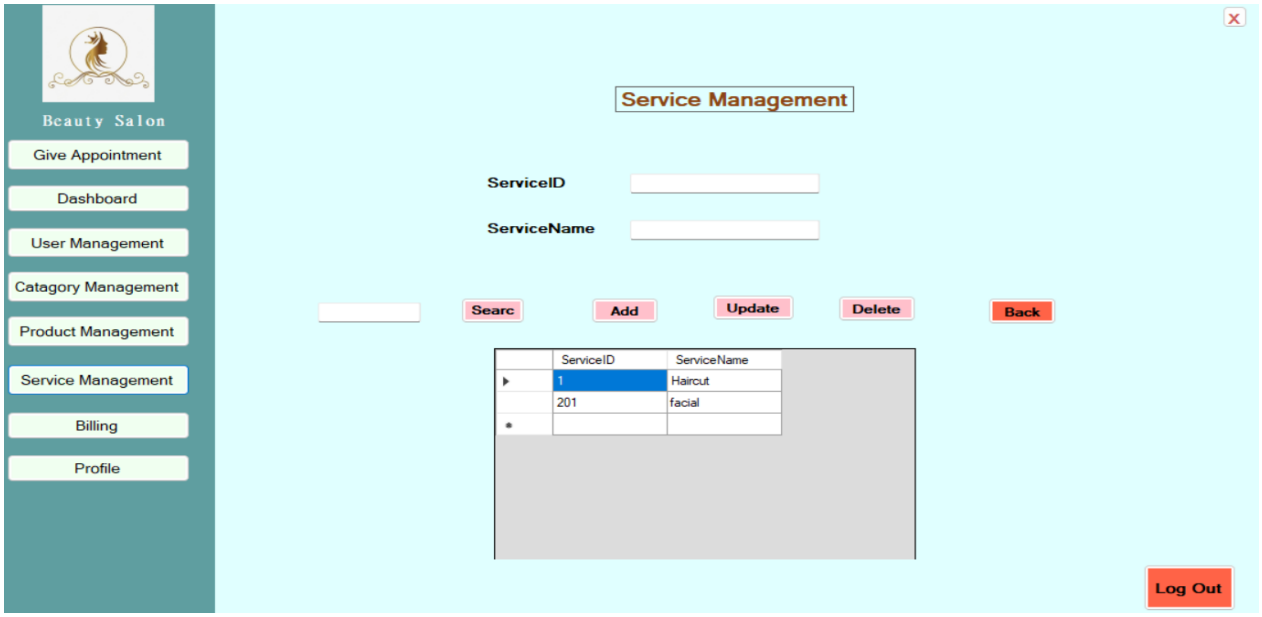
17.Category Management



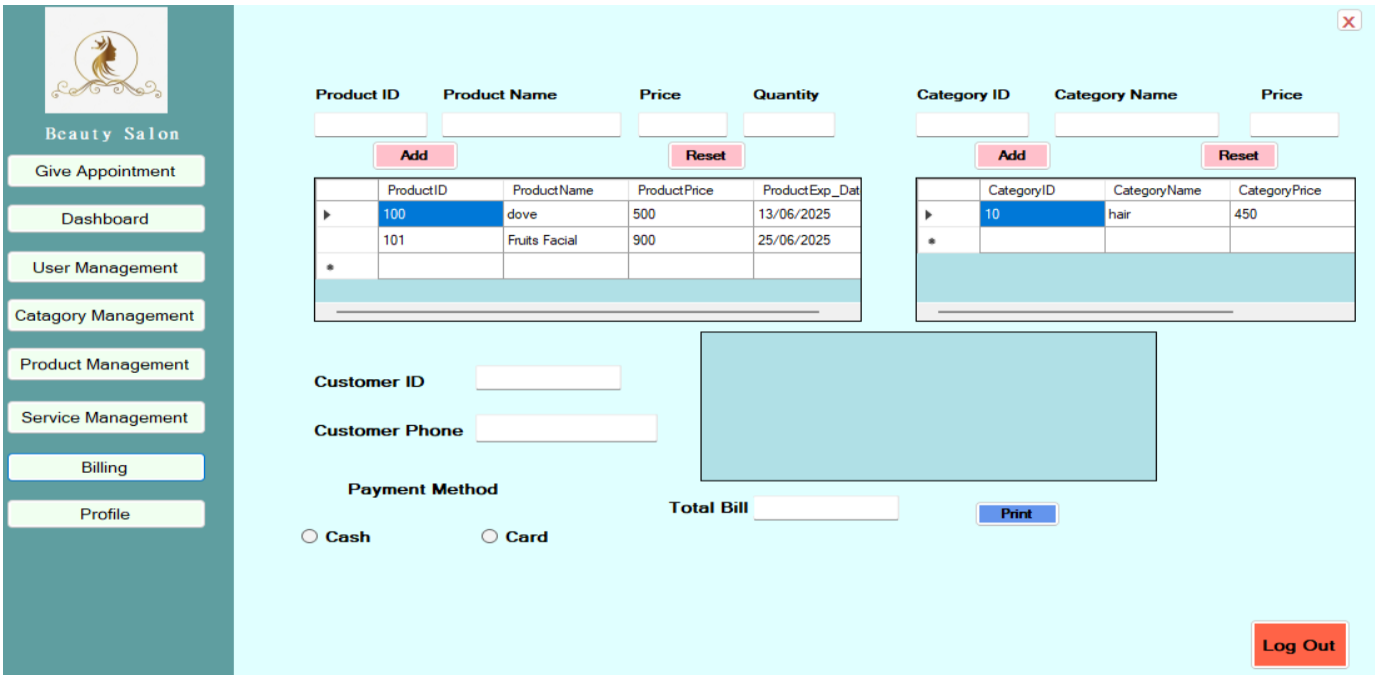
18.Product Management



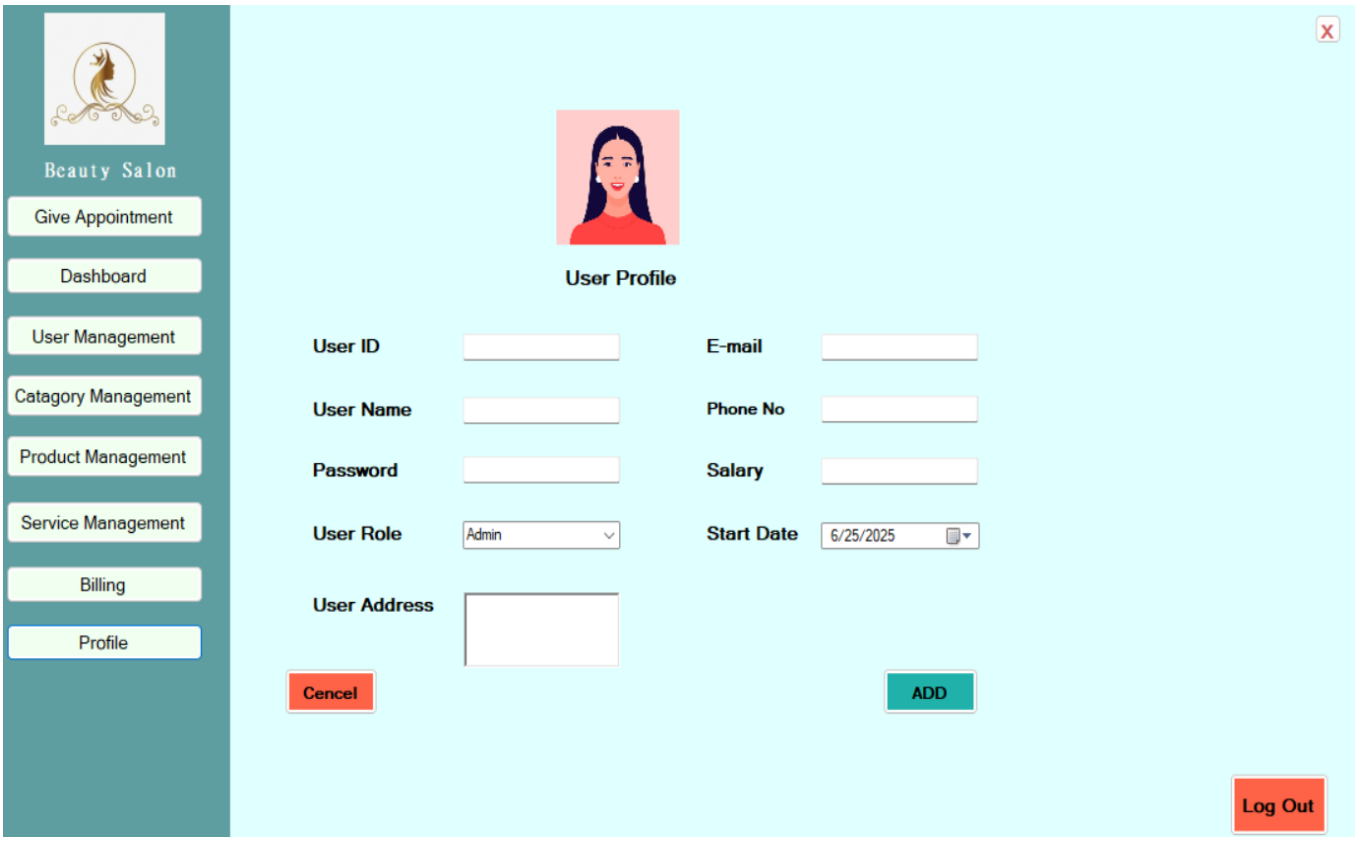
19. Service Management:



20.Billing



21.User Profile



GITHUB LINK

- https://github.com/sheikhchandni55/Beauty_Salon_Management_System